

Yeasir Rayhan

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<https://yeasirrayhanprince.github.io/>

Education

08/2021 - 05/2027 (exp)	Ph.D. in Computer Science Purdue University Advisor: Walid G. Aref
2018	B.Sc. in Computer Science & Engineering Bangladesh University of Engineering and Technology (BUET) Advisor: Tanzima Hashem Thesis: Efficient Scheduling of Generalized Group Trips in Road Networks

Past Employment

May '23 - Jun '23	Intern Database Software Engineer ITTIADB, Bellevue, WA
May '21 - May'23	Research Assistant Purdue University. Advisor: Walid G. Aref.
Jan '21 - Aug '21	Research Assistant Bangladesh University of Engineering and Technology (BUET). Advisor: Tanzima Hashem.

Publications

DaMoN'26	Yeasir Rayhan and Walid G. Aref. Virtual-Memory Assisted Buffer Management In Tiered Memory . In DaMoN@SIGMOD 2026.
SIGMOD'26	Yeasir Rayhan and Walid G. Aref. P-MOSS: Scheduling Main-Memory Indexes Over NUMA Servers Using Next Token Prediction . In ACM SIGMOD 2026.
aiDM '25	Yeasir Rayhan and Walid G. Aref. Exploring Next Token Prediction For Optimizing Databases . In aiDM@SIGMOD 2025.
SIGMOD '25	Libin Zhou, Lu Xing, Yeasir Rayhan , Walid G. Aref. GTX: A Write-Optimized Latch-free Graph Data System with Transactional Support . In ACM SIGMOD 2025.
SIGSPATIAL '23	Yeasir Rayhan and Walid G. Aref. SIMD-ified R-tree Query Processing and Optimization . In Proceedings of the 31st ACM International Conference on Advances in Geographic Information Systems.
EDBT '23	Yeasir Rayhan , Tanzima Hashem, Muhammad Aamir Cheema, Hua Lu, Mohammed Eunus Ali. An Efficient Approach for Indoor Facility Location Selection . In Proceedings of the 26th International Conference on Extending Database Technology.
TSAS '23	Yeasir Rayhan and Tanzima Hashem. AIST: An Interpretable Attention-Based Deep Learning Model for Crime Prediction . In Volume 9 of ACM Transactions on Spatial Algorithms and Systems.
TSAS '19	Yeasir Rayhan , Tanzima Hashem, Roksana Jahan, and Muhammad Aamir Cheema. 2019. Efficient Scheduling of Generalized Group Trips in Road Networks . In Volume 5 of ACM Transactions on Spatial Algorithms and Systems.

Preprints

Arxiv '26	Abdullah Al-Mamun, Yasir Rayhan and Walid G. Aref. iFVS: Towards Instance-Optimized Filtered Vector Search.
Arxiv '26	Yasir Rayhan and Walid G. Aref. Gen-DBA: Generative Database Agents .
Arxiv '26	Syed Muhammed Abubaker, Nada M. Elkordi, Yasir Rayhan , Ahmed R Mahmood, Walid G. Aref. Spatio-Temporal Access Methods: A Survey (2017-2025)-Classic, Learned, and Contact Tracing Indexes
Arxiv '25	Yasir Rayhan and Walid G. Aref. Revisiting Page Migration for Main-Memory Database Systems .
Arxiv '22	Walid G. Aref, Ahmed M. Aly, Anas Daghistani, Yasir Rayhan , Jianguo Wang, Libin Zhou. ILX: Intelligent "Location+X" Data Systems .

Talks

Purdue CS Grad Symposium'25	Profile-Guided Learned Database Kernels
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Honors

2025	Purdue Graduate Teaching Award
2025	Purdue Computer Science Graduate Student Association Travel Grant (SIGMOD '25)
2023	Travel Award, ACM SIGSPATIAL
2019	Regional Winner (Asia), The Global Undergraduate Awards (Computer Science)

Teaching

Fall, Spring '25	Teaching Assistant, Purdue University CS348: Information Systems
Fall, Spring '24	Teaching Assistant, Purdue University CS348: Information Systems
Fall '23	Teaching Assistant, Purdue University CS448: Introduction To Relational Database Systems
Spring '22	Teaching Assistant, Purdue University CS 180: Problem Solving And Object-Oriented Programming
Fall '21	Teaching Assistant, Purdue University CS 180: Problem Solving And Object-Oriented Programming

Skills

PL	C/C++, Python , Java
ML Framework	PyTorch, PyTorch Geometric